

BPG-18 · BEST PRACTICE GUIDES

Interface and subcontractor controls

Controlling the work you do not do yourself

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

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— Summary

Most of the scope on a large project is delivered by organisations that do not report to you and whose data arrives on their calendar, not yours. This guide covers the interface register and why an interface is not a milestone, the reporting cadence mismatch between a subcontractor's period end and your cut-off, keeping subcontract obligations back-to-back with the head contract, the difference between progress claimed and progress verified, and what to do when a subcontractor's own controls are too weak to produce the data your reporting depends on. The worked example separates one subcontractor claim into a payment decision and a cost decision, and shows a swing of 0.065 on the cost performance index that came from a cadence mismatch alone.

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Interface and subcontractor controls

In one paragraph. Most of the scope on a large project is delivered by organisations that do not report to you and whose data arrives on their calendar, not yours. This guide covers the interface register and why an interface is not a milestone, the reporting cadence mismatch between a subcontractor's period end and your cut-off, keeping subcontract obligations back-to-back with the head contract, the difference between progress claimed and progress verified, and what to do when a subcontractor's own controls are too weak to produce the data your reporting depends on. The worked example separates one subcontractor claim into a payment decision and a cost decision, and shows a swing of 0.065 on the cost performance index that came from a cadence mismatch alone.

Who this is for. Project controls managers and cost engineers on projects with substantial subcontracted scope; commercial managers and quantity surveyors who certify subcontractor applications; and planners managing interfaces between contractors, clients and third parties.

— 1. The control problem when the work is somebody else's

Direct work gives a controls function primary data: your timesheets, your goods-received notes, your progress measurement, on your cut-off. Subcontracted work gives you reported data — a claim prepared by an organisation with a commercial interest in the number, covering a period that ends when their accounting month ends, measured against rules they applied.

Three properties follow, and every failure in this area is a version of one of them.

The data is an assertion until you verify it. An application is a starting position in a commercial conversation, not a measurement. Treating it as one transfers the subcontractor's optimism straight into your cost report and, through it, into your forecast.

The data arrives on their calendar. Their period end and your cut-off are different dates with usually no contractual reason to converge. The gap is not rounding; it is a systematic mismatch between numerator and denominator that recurs every month in the same direction.

Their control weakness becomes your reporting weakness. No clause converts data you never received into a number you can report. You will either measure their work yourself or report an estimate, and the honest choice is to decide which in advance rather than discover it at cut-off.

— 2. The interface register, and why an interface is not a milestone

An **interface** is a point where a deliverable, information, an access, a service or a physical connection passes between two organisations. Interfaces are where integrated projects fail, because both sides believe they are managing them and neither owns the gap.

The register that manages them is not the milestone list. A milestone has one owner and one date. **An interface has two owners, two schedules and a tolerance**, and neither party can deliver it alone. That distinction determines the columns:

Field	Why it exists
Interface reference	A permanent identifier that both parties quote; never renumbered
Parties	The two organisations, named — not "us and them"
Direction	Who provides, who receives; an interface with no direction is two interfaces
What crosses	The specific deliverable, information set, access or connection, defined tightly enough to be checked
Acceptance criterion	What the receiving party must be able to do with it before it counts as delivered
Required-by date	The receiver's need date, taken from their schedule
Committed date	The provider's date, taken from theirs
Float between them	Required-by minus committed; the interface's health, in days
Owner — provider side	A named individual
Owner — receiver side	A named individual
Consequence of late	What it costs, in days or money, expressed against the receiver's critical path
Status and last confirmed	When both parties last agreed the dates, not when the register was edited

Two columns do the real work. **Float between the two dates** turns an interface from a status into a number, and a register sorted by that number is a management agenda. **Consequence of late** is what makes the provider's organisation care; an interface with no stated consequence is a request.

Bilateral confirmation is what keeps the register alive. A date taken from one party's schedule and never confirmed by the other is a hope with a date field. Confirm both sides on the record, and flag any interface not jointly confirmed within an agreed period regardless of how comfortable its float looks — comfortable float is exactly the condition under which nobody checks.

Interfaces with third parties under no contract — utilities, authorities, neighbouring projects, the client's own operations teams — need the same treatment and are most often missing from the register entirely. With no commercial lever behind them, the only control is early identification, long lead times and explicit contingency. [BPG-17 – Quantitative schedule risk analysis](#) covers what the convergence of several such paths does to a completion date, and the answer is usually worse than the schedule shows.

— 3. Cadence mismatch: their period end and your cut-off

Subcontractors report on their own commercial cycle: typically a claim to a fixed day of the month — the twentieth, say — valued and submitted a few days later, while your cost cut-off is the last day of the month and your progress measurement is taken then.

If the claim is posted unadjusted, the month's cost covers work to the twentieth and the month's earned value covers work to the thirtieth. The cost performance index — earned value divided by actual cost — then compares thirty days of output with twenty days of input, and it flatters. Next month the missing ten days arrive on top of a full month and the index drops for a reason unrelated to performance. The result is a sawtooth that costs the controls function its credibility, because everyone can see it and the explanation sounds like an excuse.

Three responses, of which only one is generally right.

Move their cut-off. Contractually the cleanest: require the subcontract's valuation date to coincide with the project's cut-off. Negotiate it at award, when it costs nothing; it is usually impossible to change later. Where their accounting calendar makes this impractical, require a progress return on your cut-off, separate from the commercial application.

Accrue the gap. Where the dates cannot be aligned, estimate the value of work performed between their period end and your cut-off and post it as an accrual, so cost and earned value describe the same period. [BPG-07 – Accruals and cut-off discipline](#) owns the accounting treatment; the refinement that belongs here is that **the gap is estimated on the verified run rate, not the claimed one**, because the claimed rate imports the over-claim into the estimate and compounds it.

Do nothing and explain it every month. Common, and not a control. If it really is the only option, publish the mismatch on the face of the report — "cost to 20th, progress to 30th" — so readers do not attribute the pattern to performance.

The same mismatch runs the other way on schedule. A subcontractor's progress update is stale by the time it enters your integrated schedule, and their float is calculated against their own network. An activity with ten days of float in their schedule may have none in yours, because yours contains the successor that matters.

— 4. Back-to-back obligations, and the gaps that are not

"Back-to-back" describes a subcontract whose obligations mirror the head contract's, so the main contractor does not carry an obligation upstream that it cannot pass down. The concept is straightforward; the failures are in the details, and they are all arithmetic on dates and percentages.

Notice regimes. If the head contract requires notice of a delaying event within a stated period and the subcontract allows a longer one, there is a window in which you can be time-barred upstream before the subcontractor is obliged to tell you anything. The exposure is the difference between the two periods, and it is worth computing for each subcontract rather than assuming the drafting handled it. Notice periods, time bars and their consequences vary substantially between contract forms and jurisdictions; the arithmetic of comparing two periods does not.

Retention. Percentages and the triggers for each release should mirror. Where the subcontract releases retention earlier than the head contract, the main contractor funds the difference. [BPG-13 – Cash flow forecasting](#) covers the working-capital consequence.

Payment timing. Paying subcontractors on shorter terms than you are paid on is a working-capital leak whose size is work in progress multiplied by the difference in days. It may be a deliberate commercial decision — it often should be — but a decision with a number attached, not an accident of two separately negotiated documents. Statutory payment regimes constrain what may be agreed in many jurisdictions and they differ; the principle of pricing the gap does not.

Change and variation mechanics. If the head contract requires a variation to be instructed and valued one way and the subcontract permits another, changes will arrive downstream that cannot be passed upstream. [BPG-11 – Change orders and variations](#) owns the change process; the point here is that the two processes must be capable of carrying the same instruction.

The controls function's role is not to draft the contract. It is to hold a **back-to-back matrix** — one table per subcontract comparing head-contract and subcontract obligation on each dimension, with

the gap quantified in days, percentage points or currency, and an owner for each gap that cannot be closed. That table turns a legal review into a managed exposure.

— 5. Progress claimed versus progress verified

A subcontractor's application states a value of work done. Your measurement states a different one. Both may be prepared honestly; the systematic difference between them is not fraud but the ordinary consequence of a party valuing its own output.

Three separate quantities are in play, and conflating any two of them produces a reporting error:

Claimed value — the subcontractor's assertion. Never posted to the cost ledger.

Certified value — what your measurement supports, valued at contract rates, less retention and previous payments. This drives payment and, through the certificate, the accrual.

Earned value — the budgeted value of physical work performed under your rules of credit. This drives performance reporting.

Certified value and earned value are close relatives but not the same quantity, and their difference is usually informative. A valuation typically pays for materials delivered to site; rules of credit typically credit installed work only. The gap between the two is therefore a timing difference that reverses when the material is installed, and reporting it as a performance variance is a mistake the worked example illustrates. [BPG-06 – Progress measurement and rules of credit](#) owns the rules that make earned value mean something.

The verification method should be agreed at award and stated in the subcontract, because retrofitting it is a negotiation. What works, in descending order of reliability: joint measurement of installed quantities against a bill; witnessed milestone achievement against defined completion criteria; independent quantity survey on a sample with a stated sampling rule; and, last, review of the subcontractor's own records. The first three are measurements. The fourth is a review of an assertion, and should be labelled as such wherever the resulting number appears.

Two practical rules. **Verify on a sample with a rule, not on whatever looks suspicious**, because a sample chosen by suspicion cannot be extrapolated and a sample chosen by rule can. And **record the over-claim, do not just remove it** — a subcontractor whose claims run consistently above measure is a forecasting signal and, eventually, a commercial conversation. A single month's over-claim is ordinary; a trend is information.

— 6. When the subcontractor's own controls are weak

Sooner or later you will engage an organisation that is good at the work and poor at the data: no coded cost structure, progress claimed as a percentage judgement, a schedule in a spreadsheet, and a programme manager who regards all of this as your problem. It is your problem, because their scope still appears in your report.

Recognise it early. The signs are visible in the first two claims: percentages moving in round numbers, progress tracking the payment curve exactly, a schedule with no logic behind the bars, no resource information, and no answer to what the remaining work consists of.

The responses that work, roughly in order of cost:

Specify the data at award. A short data schedule in the subcontract — format, fields, frequency, cut-off date, consequence of non-delivery — costs nothing at award and is unobtainable afterwards. Specify the fields, not the tool; mandating a software package excludes capable subcontractors and does not by itself produce good data.

Measure it yourself. Take the quantity survey in-house and budget for it. This is the most common practical answer, and it should be an open resourcing decision rather than an overload absorbed quietly by one cost engineer.

Shadow-schedule the scope. Maintain your own network for their work at a level sufficient to drive your integrated schedule, updated from physical observation rather than their reporting. It is duplication, and sometimes the only way to keep the interface dates honest.

Tie data to money. Where the contract allows, make a compliant progress return a condition of a valid application. The strongest lever available, but apply it as a standing requirement from the first month: a data requirement introduced during a dispute reads as a payment tactic.

Escalate as a commercial matter, not a technical one. Weak controls raise the probability of latent overrun, unnotified delay and disputed final accounts — a risk-register entry with an owner and an exposure, not a grumble in a progress meeting.

Whatever combination is used, record **the basis of the number that reaches your report.** A cost line from your own measurement and one from a subcontractor's assertion are different qualities of information, and a report presenting them identically has lost something its readers need.

— 7. How this goes wrong

The claim is posted as the cost. No verification, no accrual for the cadence gap, no distinction between claimed and certified. Everything downstream — index, forecast, revenue recognition — inherits the subcontractor's commercial position.

The gap accrual is estimated from the claimed rate. A refinement of the same error: the overclaim is removed from the certificate and then reintroduced through the accrual, which was calculated using the claim's run rate.

Interfaces are recorded as milestones. One owner, one date, no float column, no consequence. When the date slips, both parties can demonstrate that they were waiting for the other.

Third-party interfaces are missing. Utilities, permitting authorities, the client's operations staff and neighbouring projects appear nowhere, because nobody in the delivery team has a contract with them. These are usually the longest-lead and least controllable dependencies on the project.

Back-to-back is assumed, not checked. The head contract and the subcontract were reviewed by different people at different times, and the notice-period and retention differences are discovered when a claim arrives.

Materials on site are read as a performance variance. The valuation pays for stored material, the rules of credit do not credit it, the resulting index dip is investigated as a productivity problem, and next month it reverses without explanation.

Their float is treated as your float. A subcontractor reports comfortable float against their own network. Your successor activities are not in their network.

The subcontractor's schedule is accepted rather than integrated. Received as a bar chart, filed, and not linked into the project network. The consequence appears at the first interface slip, when nobody can compute its effect.

Weak controls are managed as a relationship issue. Raised repeatedly at progress meetings, never entered as a risk, never priced, never escalated commercially — until the final account arrives with no supporting records on either side.

— 8. Worked example

Illustrative figures. Currency is USD, whole dollars unless stated. One subcontract package, viewed by the main contractor. Assumptions stated as they arise; percentages rounded to one decimal place.

8.1 The setup

Item	Value
Subcontract value	4,800,000
Programme	12 months from 1 January, planned value spread evenly
Subcontractor's period end	20th of the month
Main contractor's cut-off	Last day of the month
Retention	5 %, back-to-back with the head contract
Previously certified, net of retention	1,600,000

At 20 June the subcontractor applies for a cumulative gross value of **2,150,000**. Your quantity surveyor's measure at the same date supports **2,020,000**. Measured cumulative value at 20 March was **820,000**. Earned value for the package at 30 June, under the project's rules of credit, is **2,100,000**.

8.2 The over-claim

$$\begin{aligned}
 \text{over-claim} &= \text{claimed} - \text{measured} \\
 &= 2,150,000 - 2,020,000 \\
 &= 130,000
 \end{aligned}$$

$$\text{as a proportion of the claim} = 130,000 \div 2,150,000 = 6.0 \%$$

8.3 The payment decision — the certificate

$$\begin{aligned}
 \text{net certified to date} &= \text{measured} \times (1 - \text{retention}) \\
 &= 2,020,000 \times 0.95 \\
 &= 1,919,000
 \end{aligned}$$

$$\begin{aligned}
 \text{this certificate} &= \text{net certified to date} - \text{previously certified net} \\
 &= 1,919,000 - 1,600,000 \\
 &= 319,000
 \end{aligned}$$

Had the application been certified as submitted:

$$\begin{aligned} 2,150,000 \times 0.95 &= 2,042,500 \\ 2,042,500 - 1,600,000 &= 442,500 \\ \text{difference} &= 442,500 - 319,000 = 123,500 \end{aligned}$$

which reconciles to the over-claim net of retention: $130,000 \times 0.95 = 123,500$. Certifying the application rather than the measure would have paid **123,500** more than the work supports, this month.

8.4 The cost decision — closing the cadence gap

The certificate covers work to 20 June. The cost report closes on 30 June. The gap is **10 days**.

The run rate is taken from verified measure over the three preceding periods, not from the claims:

$$\begin{aligned} \text{measured growth} &= 2,020,000 - 820,000 = 1,200,000 \quad (20 \text{ March to } 20 \text{ June}) \\ \text{per month} &= 1,200,000 \div 3 = 400,000 \\ \text{per day (30-day)} &= 400,000 \div 30 = 13,333.33 \\ \text{gap accrual} &= 13,333.33 \times 10 = 133,333 \end{aligned}$$

Cost to be recognised at 30 June for this package:

$$2,020,000 \text{ (certified measure)} + 133,333 \text{ (accrued 21–30 June)} = 2,153,333$$

A coincidence worth noticing, because it is a trap. The number arrived at — 2,153,333 — is within 3,333 of the 2,150,000 the subcontractor claimed. A cost engineer who had simply posted the claim would have been almost exactly right this month, which is precisely why the shortcut survives. But the two figures are right for different reasons and are composed differently: 2,153,333 is 2,020,000 of verified work plus 133,333 of estimated work, which supports three separate actions — certify 319,000, accrue 133,333 with no payment attached, and pursue the 130,000 over-claim. The single figure of 2,150,000 supports none of them, and the agreement between the two numbers will not survive any change in the over-claim or the run rate.

8.5 What it does to the reported indices

Assumption: the package's planned value is spread evenly across twelve months from 1 January, so planned value at 30 June is six months of 400,000. SPI is the schedule performance index, earned value divided by planned value.

$$\begin{aligned} \text{PV (planned value)} &= 6 \times 400,000 = 2,400,000 \\ \text{EV (earned value)} &= 2,100,000 \\ \text{AC (actual cost)} &= 2,153,333 \\ \text{SPI} &= \text{EV} \div \text{PV} = 2,100,000 \div 2,400,000 = 0.875 \\ \text{CPI} &= \text{EV} \div \text{AC} = 2,100,000 \div 2,153,333 = 0.975 \end{aligned}$$

Now the same month with the cadence gap ignored — certificate posted, no accrual:

$$\begin{aligned} AC &= 2,020,000 \\ CPI &= 2,100,000 \div 2,020,000 = 1.040 \end{aligned}$$

$$\text{swing} = 1.040 - 0.975 = 0.065$$

A swing of 0.065 on the cost performance index, produced entirely by a ten-day mismatch between two calendars. On a tolerance band a few hundredths wide, that is the difference between green and an exception report — and next month, when the missing ten days land on top of a full period, the index will fall for a reason that has nothing to do with how the work was performed.

8.6 Reading the 0.975 correctly

The 0.975 should not be reported as a productivity problem. The recognised cost of 2,153,333 is a valuation basis; the earned value of 2,100,000 is a rules-of-credit basis. The difference:

$$2,153,333 - 2,100,000 = 53,333$$

is material delivered to site and paid for, which the valuation includes and the rules of credit do not credit until it is installed. It is a timing difference that reverses on installation, and the narrative should say so. Investigating it as a performance variance would consume a month of attention and find nothing, which is a real cost even though it does not appear in any ledger.

8.7 The schedule reading

The package is behind, and that reading is unaffected by any of the cost treatment above:

$$\begin{aligned} \text{measured completion} &= 2,020,000 \div 4,800,000 = 42.1 \% \\ \text{planned completion at 30 June} &= 2,400,000 \div 4,800,000 = 50.0 \% \\ \text{SPI} &= 0.875 \end{aligned}$$

A package at a schedule performance index of 0.875 with a 6.0 per cent over-claim on its latest application is a specific management situation: the subcontractor is behind and is valuing optimistically, which is the ordinary sequence and the moment at which the interface dates downstream of this package should be re-confirmed with the parties that depend on them.

9. Checklist

At award

- [] Valuation date aligned to the project cut-off, or a separate progress return required on the cut-off.
- [] Data schedule in the subcontract: fields, format, frequency, cut-off, consequence of non-delivery.
- [] Verification method stated — joint measurement, witnessed milestone, sampled survey or record review.

- [] Back-to-back matrix completed: notice periods, retention percentages and triggers, payment timing, change mechanics — each gap quantified in days, points or currency, each with an owner.
- [] Interface points identified, including third parties with no contract.
- [] Rules of credit for the package agreed and written, distinct from the valuation basis.

Every period

- [] Claim measured before it is certified; over-claim recorded, not just removed.
- [] Certificate computed from measure, retention and previous payments — with the arithmetic shown.
- [] Cadence gap accrued, using the verified run rate.
- [] Cost basis labelled: measured, certified, accrued or asserted.
- [] Materials on site identified and excluded from the performance narrative.
- [] Subcontractor's progress update integrated into the project network, not filed.
- [] Interface register updated with dates confirmed by both parties, and float recomputed.
- [] Over-claim trend reviewed across periods, not only this month's figure.

Every quarter, or at each gate

- [] Interfaces with no bilateral confirmation in the agreed period flagged regardless of float.
- [] Subcontractors with weak controls carried as named risk entries with an exposure, not as grievances.
- [] Back-to-back gaps that could not be closed reviewed for whether they have been triggered.
- [] The cost of measuring subcontracted work in-house reviewed against the resource actually assigned.

The projects that lose control of subcontracted scope rarely do so through one large failure. They do it through a claim posted as a cost, an interface with one owner and a notice period that was seven days out of step — each individually small, each invisible in the report, and all of them arriving together in the final account.

— Related

- [BPG-06 – Progress measurement and rules of credit](#) — the rules that make earned value on subcontracted scope mean something
- [BPG-07 – Accruals and cut-off discipline](#) — the accounting treatment behind the gap accrual in §8.4
- [BPG-11 – Change orders and variations](#) — the change process that must be capable of carrying an instruction in both directions
- [BPG-12 – Claims and extension of time](#) — where notice-period gaps and unpassed subcontractor claims end up
- [BPG-13 – Cash flow forecasting](#) — the working-capital effect of payment and retention terms that are not back-to-back
- [BPG-05 – Schedule quality – a practical review](#) — the checks to apply to a subcontractor's schedule before integrating it

— Sources and standards

- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 7 — Contracts, Commercial Management, BoQ, Invoicing and Revenue, first authored draft, August 2026: interim valuation and certification, back-to-back subcontract discipline, and the reconciliation of earned value, valuation and revenue.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 5 — Cost Management and Cost Control, first authored draft, August 2026: commitment and accrual capture, and the cost ledger's relationship to certification.
- PCL-AI Body of Knowledge ([docs/bok/](#)), Domain 10 — Project Scheduling, first authored draft, August 2026: integration of contractor schedules and the treatment of float across networks.
- PCI Canonical Facts ([docs/publication-framework/00-framework/CANONICAL-FACTS.md](#)), verified August 2026: naming, status and claims policy.

Notice periods, time bars, retention mechanics, payment terms and statutory payment regimes vary between contract forms and between jurisdictions. This guide names no contract form and reproduces no clause; where a mechanism is described it is described generically, and the reader is expected to apply the arithmetic to the documents actually in force on their project.

— Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.